

If you're using a remote server then to further increase security we can set up SSH Public key authentication. SSH or Secure Shell is an encryption-based method to remotely operate your servers.

First log in as the appropriate user

- `$ su - new_user`

Generate the RSA key

- `$ ssh-keygen`

The output screen will look something similar to this:

- `$ Output`
- `Generating public/private rsa key pair.`
- `Enter file in which to save the key (/your_home/.ssh/id_rsa):`

Press enter to save it in your home directory's .ssh subfolder.

You will get a prompt to secure with an additional password. Choose a strong password and press 'enter' to proceed.

- `$ ssh-copy-id new_user@remote_server`

This will install your SSH key on the remote server given that you have the proper credentials to log into the server which will be asked of you during the installation.

If the installation was successful you'll get a confirmation message on terminal.

Now whenever you want to log into the server, a short, one line of command will suffice.

- `$ ssh username@remote_host`

Now that you've installed SSH keys, it's better to disable the normal password login.

First step is logging into your server using the SSH keys we just generated.

- `$ sudo nano /etc/ssh/sshd_config`

Within that file search for password authentication and set it to 'no'. Save and restart the SSH process.

- `$ sudo systemctl restart ssh`

Configure your firewall

SSH protects your server from brute-force attacks and unwanted access but you're still vulnerable over the network level. Ubuntu uses UFW (Uncomplicated Firewall) as its default firewall.

To check the application status, open your terminal and issue the following commands.